Universitat Jaume I

Bachelor's Thesis in Video Game Design and Development

Micorrizas

Design and development of a geolocated mobile video game to promote knowledge and relationship with the natural environment

Student	Saúl Pacheco Trilles
Supervisor	Águeda Gómez Cambronero
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ABSTRACT

This document presents the Bachelor's Thesis developed for the Bachelor's Degree in Video Game Design and Development. The project originates from a pedagogical activity carried out with Early Childhood Education students in the course MI1815 – The Natural Environment in Early Childhood Education, consisting of a guided visit to the *Jardín de los Sentidos*, an outdoor space at Jaume I University characterised by its rich biodiversity. The educational value of this activity lies in encouraging students to observe, discuss, and reflect on natural ecosystems through direct interaction with the environment. Building upon this experience, this project explores how educational objectives of the original activity can be transferred to a digital, autonomous and playful format. To address this challenge, *Micorrizas* was designed as an educational video game that promotes environmental awareness and engagement through active exploration and collaborative learning. The resulting game is a cooperative location-based mobile experience developed with Unity, where players interact with real-world locations while progressing through shared challenges and decision-making processes. To support this experience, the game combines geolocation and multiplayer technologies that enable collaborative gameplay in outdoor environments. The game has also been evaluated by the lecturers responsible for the original educational activity, providing feedback on its suitability as a tool for environmental education in outdoor learning contexts.

KEYWORDS

Serious game - educational game - cooperative multiplayer - geolocation - mobile learning - environmental education - Jardín de los Sentidos.

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CHAPTER 1

INTRODUCTION

1.1. CONTEXT

Environmental education is increasingly recognised as an essential component of sustainable development, as it promotes the knowledge, attitudes, and competencies required to understand and address environmental challenges. UNESCO highlights the importance of integrating sustainability and climate action into educational practices, promoting learning experiences that encourage critical thinking and environmental responsibility [1]. In this context, direct contact with nature has been shown to support environmentally responsible attitudes and behaviours, particularly when learners engage in active and meaningful experiences [2].

One example of this approach is the educational activity carried out in the subject MI1815 - The Natural Environment in Early Childhood Education at Jaume I University. The activity takes place in the *Jardín de los Sentidos*, located on the Riu Sec campus and recognized as one of the university's three main botanical areas of interest. The garden is a botanical and cultural space of approximately 1,500 m², organised into six thematic platforms: a welcome area and five areas related to sensory perception. It contains more than 260 plant species belonging to 78 different families. Established in 2004 as both an educational and cultural resource, the garden serves as an outdoor laboratory where future teachers can engage directly with the natural environment.

Through guided exploration of the garden, students observe a wide variety of plant species from different regions of the world, develop an appreciation for botanical heritage, and learn to interpret ecological relationships within a real-world context. The sensory-oriented design of the garden also provides opportunities to connect botanical knowledge with pedagogical practices relevant to early childhood education. In addition, the activity encourages students to design and reflect on educational itineraries in natural settings, highlighting the value of nearby green spaces as accessible learning environments that can support environmental education when access to larger natural areas is limited.

1.2. MOTIVATION

Although the *Jardín de los Sentidos* activity provides valuable educational experiences, there are currently several limitations to its impact. As it takes place outdoors and within a specific academic timeframe, its implementation is constrained by the duration of the session, environmental conditions, group availability, and the presence of an expert instructor capable of guiding observations, answering questions, and adapting the experience to the learners' needs. These constraints make it difficult to repeat the activity frequently, extend it to additional groups, or maintain it as an ongoing learning process. Consequently, there is a need to explore alternative approaches that preserve the experiential value of the activity while enabling more autonomous, flexible, and sustained participation.

In this context, serious games and mobile technologies offer a promising approach for extending such learning experiences into more accessible, interactive, and engaging educational formats. Serious games are digital games designed with a main purpose that goes beyond entertainment. Although they preserve essential elements of play, such as rules, challenges, interaction and feedback, their design is oriented towards educational, training, social or behavioural objectives [3, 4].

In educational contexts, serious games are especially relevant because they can place learners in an active role. Instead of receiving information passively, players interact with content, make decisions, test hypotheses and progressively construct meaning through experience. This approach is closely related to constructivist and experiential learning perspectives, as the game environment can encourage exploration, problem solving and reflection. In addition, serious games can offer safe spaces in which learners are allowed to make mistakes and observe their consequences without the risks associated with real-world practice [5, 3, 4].

Another key contribution of serious games is their capacity to provide immediate feedback. The system can respond directly to the player's actions, allowing learners to adjust their strategies, correct errors and continue progressing through an iterative process. This feedback can also support assessment, not as an external interruption of the activity, but as part of the game experience itself. However, the educational value of serious games depends on their design and integration within a broader learning context. Poor interface design, excessive cognitive load, technical problems or lack of pedagogical alignment can reduce their effectiveness. Therefore, serious games should be understood as complementary tools rather than replacements for traditional teaching methods [6, 4, 3].

The increasing interest in serious games is also linked to technological development. Recent advances in mobile devices, mixed reality, augmented reality, virtual reality, artificial intelligence and learning analytics have expanded the possibilities for creating interactive educational experiences situated in physical or simulated environments. In this context, mobile and location-based games are particularly relevant because they can connect digital interaction with real-world exploration, making the environment itself part of the learning process [5, 3].

By combining these approaches, guided outdoor educational activities can be transformed into autonomous and interactive learning experiences in which students engage with challenges, receive feedback, collaborate with peers, and construct environmental knowledge through the exploration of real-world environments.

From this perspective, the present project emerged from a collaboration between the Department of Early Childhood Education and the CEVI research group, to which the supervisor of this Bachelor's Thesis belongs. The aim is to transfer the core learning outcomes of the original activity into a mobile experience that preserves interaction with the real environment while reducing the need for continuous teacher guidance and the constraints associated with scheduled learning sessions. By integrating geolocation, collaborative gameplay and situated learning principles, the proposed solution (*Micorrizas*) aims to provide a more autonomous, scalable and engaging educational experience that complements the original activity and supports environmental education beyond the classroom.

This project is also motivated by professional and personal development. From a technical perspective, it represents a challenge because it covers areas I had not previously developed in my university projects, particularly multiplayer interaction and geolocation-based gameplay. The implementation of a serious mobile game that combines map integration, location-based triggers, cooperative synchronisation and mini-games offers the opportunity to expand my skills as a video game developer. At the same time, the project is personally motivating because it connects game design with a real educational context, allowing the final prototype to address a specific need and explore how video games can support learning.

1.3. OBJECTIVES

The main objective of this project was to design and develop an educational video game that promotes environmental awareness and engagement through active exploration and collaborative learning.

The specific objectives to achieve the main goal are the following:

1. To transform the educational activity conducted in the *Jardín de los Sentidos* into a videogame while preserving its pedagogical contents.
2. To design cooperative multiplayer gameplay mechanics that encourage exploration, observation, communication, and collaborative problem-solving.
3. To develop an Android mobile game that reacts to the player's physical position within the *Jardín de los Sentidos*.
4. To provide teachers and the students of the subject an educational resource that complements traditional guided visits and promotes autonomous learning.
5. To increase awareness and appreciation of biodiversity through interactions within the natural environment.

CHAPTER 2

PLANNING AND RESOURCES EVALUATION

2.1. PLANNING

This section presents de organisation of the project work. To facilitate its development process, the project was divided into a set of tasks, as outlined below. Trello [7] was used to manage the development process, organise pending tasks and track progress throughout the project.

- **Project analysis** (10 hours):
  - **Meetings and research** (6 hours): In order to ensure that the game would adequately meet the educational needs and align with the objectives of the training activity, a number of meetings were held with the teachers involved, both during the initial phase and throughout the course of the project. These sessions made it possible to gather ongoing feedback and guide design and development decisions. The teachers also provided material about the garden, its species and the educational activities carried out there, which required additional research in order to understand the context and adapt the game content appropriately. Furthermore, to address certain technical aspects of the project, contact was maintained with a geolocation specialist, whose advice helped to resolve specific issues relating to this area.
  - **Initial planning** (4 hours): The initial planning phase consisted of defining the scope of the project, identifying the main tasks to be carried out and estimating the time required for each development stage.
- **Game design** (50 hours): The game design phase is an ongoing process, as the project is constantly evolving. A key aspect of game design is ensuring that the focus of the serious game is not lost sight of. This includes ensuring the game is well-balanced, designing cooperative missions, and making sure the game is rewarding for the user, while meeting the pedagogical goals.
- **Art design** (10 hours): This task involved both sourcing and creating the visual and audio assets used in the game. Part of this time was spent searching for references and assets that matched the project's artistic style and narrative, primarily through platforms such as the Unity Asset Store [8] and Itch.io game-assets [9]. It also involved creating or adapting certain necessary resources, such as the character sprites, the images for the Garden of Hearing, and the image representing the mycorrhizae found above each garden. Furthermore, audio resources were explored for the Garden of Hearing quest, including testing recorded sounds and searching specialised sound databases such as Xeno-canto [10], which offers recordings of birdsong and other natural sound material.
- **Development** (120 hours): This was the longest phase of the project and the one with the most iterations. It was divided into several implementation milestones, which are detailed below, including geolocation, multiplayer mode, mission logic and final integration.
  - **Geolocation** (60 hours): Geolocation was the most complex part of the development process.

This task required an initial research phase to understand the available location services and their limitations, followed by several rounds of implementation and testing. Its complexity was compounded by the difficulty of validating GPS behaviour outside the development environment, as reliable testing had to be carried out at the actual physical location.

- ✦ **Multiplayer** (20 hours): This task involved researching multiplayer functionality and subsequently implementing the system using the Unity Relay SDK. It included configuring the network service, creating and joining game sessions, as well as ensuring the necessary data synchronisation between players.
- ✦ **Quests** (30 hours): This task focused on designing, implementing and refining the quest system. Each quest had a different narrative and pedagogical purpose, which required specific mechanics, interaction models and feedback systems.
- ✦ **Integration** (10 hours): This task involved bringing together all the previously implemented elements and ensuring they functioned as a single, coherent prototype. Specifically, this involved integrating the geolocation system, the multiplayer functionality, the mission logic and user interaction, as well as testing the application's entire workflow from start to finish in order to iterate and improve it.

✦ **Testing and iteration** (60 hours): Aquí se ha contabilizado el tiempo de testear cada mecánica de las quest y el tiempo de ir a testear la geolocalización.

✦ **Documentation** (50 hours): This task covered the preparation of the project documentation throughout the development process.

- ✦ **The technical proposal** (5 hours): The technical proposal defined the scope of the project. Its aim was to set out the basic concept of the project, the educational context, the main objectives and the initial planning. This document also helped to clarify important conceptual aspects of the project, such as the distinction between gamification and serious games.
- ✦ **GDD** (10 hours): This document defined the core gameplay loop, cooperative mechanics, narrative framework, level structure, interface design, multiplayer model and technical architecture.
- ✦ **The Dissertation** (25 hours): The final report is a more comprehensive piece of documentation; for this reason, it was planned to include the theoretical underpinnings of serious games, the rationale behind the project, design decisions, and so on. The previous documents served as the basis for this report, but the content had to be reorganised and expanded from an academic perspective.
- ✦ **Project defence** (10 hours): Finally, the presentation was left until the final phase of the project, once all design and implementation decisions had been made. The presentation was a summary of the most relevant aspects of the project.

Figure 2.1 presents the final Gantt chart of the project, including the tasks carried out and the time allocated to each of them.

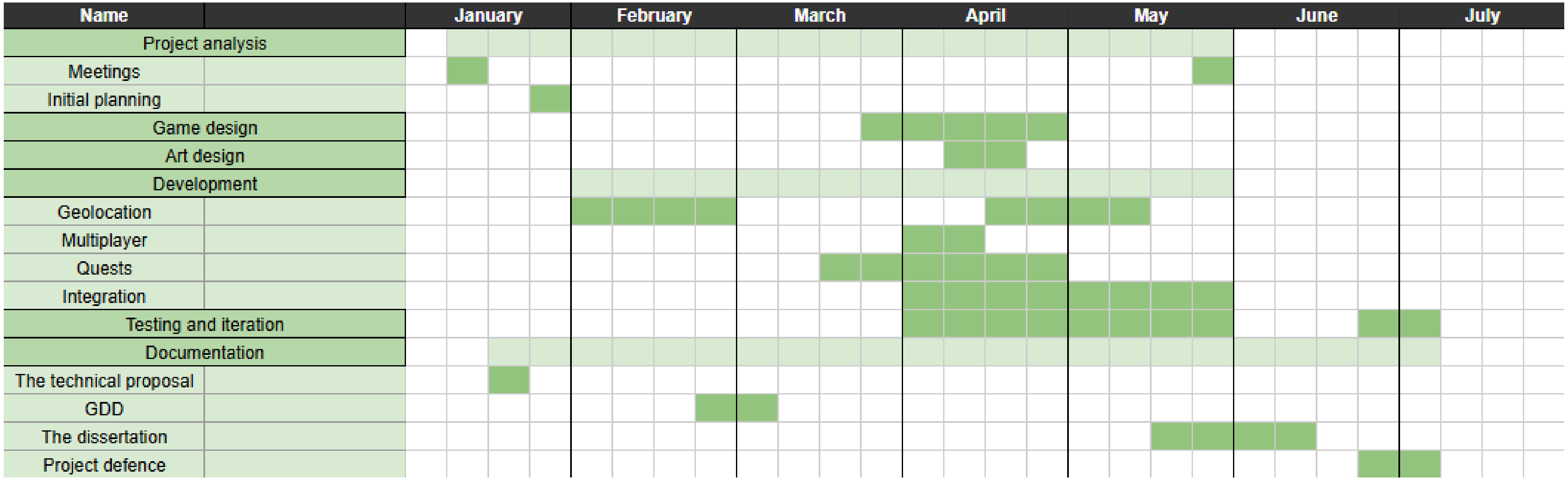

Figure 2.1: Gantt chart of the project planning

2.2. COSTS

The cost of carrying out this project is broken down below. This does not represent an actual payment, as the project was undertaken as an academic assignment, but it provides an estimate of the professional value of the work carried out

2.2.1. WORKING HOURS

The cost of the work carried out in this project has been estimated according to the number of development hours planned in the Gantt chart. To assign an economic value to these hours, the reference used was the national collective agreement for consultancy and information technology companies, published in the Spanish Official State Gazette (BOE) [11].

The selected professional category was Area 3, Group D, Level 3, since it corresponds to software development and programming tasks and represents an appropriate reference for an entry-level profile in the software industry. According to the updated 2026 salary tables, this category has an annual gross salary of 18,542.34 €. Considering the annual working time of 1,800 hours established in the same agreement, the hourly cost is calculated as follows:

$$\frac{18{,}542.34\,€}{1{,}800} = 10.30\,€/\text{hour}$$

Therefore, the estimated labour cost of the project is:

$$\text{Total labour cost} = \text{Number of hours} \times 10.30\,€/\text{hour}$$

If the project required 330 hours of work:

$$330\,\text{h} \times 10.30\,€/\text{h} = 3,399\,€$$

2.2.2. HARDWARE

The development of a similar project would require a computer capable of running the necessary development tools primarily Unity as well as several mobile devices, between two and six devices. We will estimate six devices to be able to test the application under full conditions. The hardware specifications of the system used for this project is described below.

- OMEN Gaming Laptop 16-ap0xxx [12], 32GB of RAM and 954GB of hard disk, NVIDIA GeForce RTX 5070 Laptop GPU. Cost: 1,399€.
- Xiaomi Redmi Note 13 Pro 5G [13]. Cost: 80€

In order to provide a realistic cost reference to replicate the project, two alternative scenarios can be considered: purchasing a similar device or renting additional test devices. The official Xiaomi reference price for the Redmi Note 13 Pro 5G is approximately 219.99 €. As another option, the company could rent one or two additional Android devices during the validation period. Although the exact Xiaomi Redmi Note 13 Pro 5G was not available in the consulted renting catalogue, similar Android devices were available through Rentik / Phone House [14], such as the Xiaomi Redmi Note 15C from 6.90 €/month, the Motorola G55 5G from 8.90 €/month, or the Xiaomi Redmi Note 14 5G from 16.90 €/month.

The current range is between 6.90 €/month and 16.90 €/month, so an average cost of approximately 10 €/month/device will be estimated. Considering that, during one month, only two devices may be needed for testing, but during another month, up to six devices may be required.Therefore, the estimated cost would be 20 € for the first month plus 60 € for the second month, resulting in a total of approximately 80 €.

2.2.3. SOFTWARE

The software tools and licences required to develop this project were free for educational use, but their commercial value is detailed below for replication in a professional setting:

- **Unity:** Used as the main game engine. While it currently offers a free Personal edition, the pricing structure was recently modified, cancelling the Runtime Fee [15]. For a professional studio, a Unity Pro licence might be required depending on revenue.
- **ArcGIS SDK:** Used for the map and geolocation features. A student use licence [16] is available at no cost for academic purposes, whereas commercial development requires a paid subscription.
- **OpenAI Codex:** Used to assist in the development of certain scripts and technical problem-solving. The pricing model [17] is 20 US$/month/user. The 300 hours of project development took around 4 months, so this would cost around $4 \times 20 = 80$ €.

2.2.4. OTHER EXPENSES

Since the project is based on a geolocated experience in the *Jardín de los Sentidos*, several field tests had to be carried out in the real environment to validate GPS behaviour, interaction areas and mobile usability. Therefore, travel expenses were included as an additional project cost. Each validation trip was estimated as a 15 km journey. Using the official Spanish reference of 0.26 €/km for private vehicle travel expenses, the cost per visit is: Number of visits × 15 × 0.26 = Travel cost $15 \times 0.26 = 3.9$€/visit Assuming a total of ten validation visits, the estimated travel cost would be: $10 \times 3.9 = 39$€

2.2.5. ESTIMATED TOTAL COST

After analysing the different cost categories, the estimated cost of replicating the project in a professional context amounts to 4,798 €. This total includes the estimated labour cost, the hardware required for development and testing, the software tools and licences, and the travel expenses associated with field validation. Table 2.1 summarises the final cost breakdown for each category.

Table 2.1: General project cost, broken down into the main cost categories

Concept	Cost (€)
Working hours	3,399 €
Hardware	1,479 €
Software	80 €
Other expenses	39 €
Total	**4,997 €**

2.3. PLANNING DEVIATIONS

Table 2.2: Planning deviations

Task	Estimated hours	Hours worked	Deviation
Project analysis	15	10	-5
Game design	80	50	-30
Art design	15	10	-5
Development	130	150	+20
Testing and iteration	30	60	+30
Documentation	30	50	+20
Total	**300**	**330**	**+30**

The project presented several deviations from the initial planning. As shown in Table 2.2, the final workload exceeded the initial estimation by 30 hours. Table 2.2 shows the comparison between the initially estimated hours, the hours finally spent on each task, and the resulting deviation.

The tasks that required less time were project analysis, game design and art design. In the case of project analysis, the scope of the project and the main requirements were defined at an early stage, mainly thanks to the initial meetings with the teachers involved in the educational activity. Game design also required fewer hours than initially planned because some design decisions were simplified during development in order to keep the prototype feasible and focused on the main educational objectives. Similarly, art design took less time than expected because several visual resources were obtained from existing asset libraries or adapted from available material, reducing the need to create every asset from scratch.

In contrast, development, testing and documentation required more time than originally estimated. Development exceeded the initial planning because the implementation of geolocation and the integration of the different systems involved more technical complexity than expected. The project required several iterations to connect the map, GPS position and mission flow, since the application could not be validated inside the Unity editor. The geolocation system had to be tested on Android devices and in the real physical environment of the *Jardín de los Sentidos*.

The project was completed in 330 hours instead of the 300 hours initially estimated. The final deviation of 30 hours was mainly caused by the complexity of developing and validating a geolocated mobile application in a real outdoor environment.

CHAPTER 3

ANALYSIS

This chapter identifies the main requirements that guided the design and development of *Micorrizas*. The purpose of this analysis is to establish what the system must provide in order to support the intended experience.

The requirements were defined from three main sources: the educational needs of the activity, the game design decisions and the technical limitations of mobile development. The educational context required the game to support biodiversity learning through observation and exploration. The game design required cooperative interaction, narrative progression and feedback. The technical context required geolocation, map integration, multiplayer synchronisation and usability in an outdoor environment.

The requirements are divided into functional (Table 3.1) and non-functional requirements (Table 3.2). Functional requirements describe the actions and behaviours that the system must implement. Non-functional requirements describe the quality conditions that the system must satisfy during use.

3.1. FUNCTIONAL REQUIREMENTS

Table 3.1: Functional requirements

ID	Description
FR01	**Create a game session** The system must allow a player to create a multiplayer session that other users can join.
FR02	**Join a game session** The system must allow players to join an existing session using a code.
FR03	**Register player names** The system must allow players to identify themselves within the group session.
FR04	**Display the game map** The system must show the map of the *Jardín de los Sentidos* using the selected map integration.
FR05	**Show the player real position** The system must display the approximate position of the player on the map.
FR06	**Detect sensory areas** The system must detect when the player enters the activation area of a sensory garden.

Continued on next page

ID	Description
FR07	**Activate quests by location** The system must unlock the corresponding quest when the player reaches the required area.
FR08	**Present narrative content** The system must display narrative dialogue related to the activated area and the current state of the experience.
FR09	**Include one quest per sensory garden** The system must include specific quests for sight, sound, touch, taste and smell.
FR10	**Support cooperative gameplay** The system must require players to collaborate during the quests.
FR11	**Synchronise session state** The system must keep all connected devices updated with the active quest, progress and results.
FR12	**Allow answer submission** The system must allow players to submit answers during the quests.
FR13	**Calculate user performance** The system must calculate a group score after each quest.
FR14	**Calculate the final result** The system must calculate and display the final group result after all required areas have been completed.
FR15	**Provide feedback** The system must provide feedback after each challenge, indicating the result of the group's action.
FR16	**Track completed areas** The system must register which sensory gardens have already been completed.
FR17	**Display final summary** The system must present a final screen with the group members and their final score.
FR18	**Allow replay or restart** The system should allow the group to restart the experience if required.

3.2. NON-FUNCTIONAL REQUIREMENTS

Table 3.2: Non-functional requirements

ID	Description
NFR01	**Mobile platform** The application must run on Android mobile devices.

Continued on next page

ID	Description
NFR02	**Outdoor readability** The interface must remain readable in outdoor conditions, including sunlight and variable lighting.
NFR03	**GPS tolerance** Activation areas must be wide enough to compensate for inaccuracies in mobile GPS positioning.
NFR04	**Performance** The application must avoid unnecessary graphical complexity and maintain stable performance on mobile devices.
NFR05	**Loading efficiency** The system should minimise waiting times during map loading, scene transitions and quest activation.
NFR06	**Small group support** The multiplayer system must support small cooperative groups.
NFR07	**Synchronisation consistency** All devices in the same session must receive the same challenge state and result updates.
NFR08	**Simple access** The system should allow players to access a session through a simple code-based flow.
NFR09	**Data minimisation** The application should only handle the information required for the session, such as player names, group progress and score.
NFR10	**Clear interaction** The interface must use simple controls, short instructions and clear feedback.
NFR11	**Maintainability** The content structure should allow new plants, sounds, questions or quests to be added in future versions.
NFR12	**Accessibility support** Audio-based information should include visual or textual support when necessary.
NFR13	**Robustness** The application should remain usable when minor GPS variations or temporary connection delays occur.
NFR14	**Battery awareness** The application should avoid excessive use of continuous processes that could increase battery consumption.

CHAPTER 4

DEVELOPMENT

4.1. GAME DESIGN

This section describes the design of *Micorrizas*, presenting the main elements that define the game and support its educational objectives. Specifically, it introduces the game concept, narrative, gameplay loop, game scenes, mechanics, and the principal design decisions adopted during the development process.

4.1.1. GAME CONCEPT

Micorrizas is a multiplayer (2-6) cooperative location-based game in which players move through real-world locations shown on a map, triggering quests that consist of short collaborative challenges with varied mechanics.

Micorrizas is based on the educational activity carried out in the *Jardín de los Sentidos* as part of the subject related to the natural environment in Early Childhood Education. In this activity, students separated in groups visit the garden and work directly with the natural space through observation, identification and interpretation tasks. The aim is for students to understand the educational potential of nearby natural environments and to learn about the biodiversity of the garden through direct interaction with plants, sensory areas and ecological elements.

Since the original activity takes place outdoors and requires students to move through the different areas of the garden, *Micorrizas* was designed as a mobile and geolocated experience. The mobile device makes it possible to guide the students through the physical space, detect their position and activate missions when the group reaches specific areas. In this way, the game preserves the relationship with the real environment while adding a digital layer of interaction, narrative progression and feedback.

The content of the game is organised through missions linked to the senses represented in the garden (see Figure 4.1). Each quest focuses on a specific type of perception or knowledge: visual recognition of plants, identification of sounds, deduction of species from tactile characteristics, the relationship between fruits and species, and classification of plants according to their uses. The design of each quest was defined by combining the educational purpose of the original activity by each area of the garden (i.e., Garden of Sight, Garden of Hearing, Garden of Touch, Garden of Taste, Garden of Smell) with interaction models suitable for mobile devices. This type of interaction is detailed in Section 4.1.5.

Figure 4.1: The *Jardín de los Sentidos*, organised into its sensory zones

Cooperation plays a central role in the design. Quests are completed as a group and the score obtained belongs to all members in the same game. Some quests rely on discussion and collective memory, while others distribute information across devices to encourage communication between players. This cooperative gameplay makes collaboration an active part of how the game functions and reflects the social nature of the original learning activity.

Main character's narrative and the mycorrhizal network give unity to the experience. Each mission represents the recovery of a part of the garden and contributes to rebuilding the lost connection between the sensory zones. The game's progression is presented as a symbolic restoration of the natural environment, linking players' actions to the observation, interpretation and care of biodiversity.

4.1.2. LORE

Deeproot (see Figure 4.2) is the main character in *Micorrizas* and acts as the game's narrative guide. He is an Ent, a creature directly linked to the flora of the *Jardín de los Sentidos*, whose role is to protect and manage the area's natural resources. Due to the slowness inherent to his species and the progressive deterioration of the garden, Deeproot seeks the help of a group of adventurers, represented by the players, to restore the lost balance.

Figure 4.2: Deeproot the main character

The story begins with a conflict caused by the negligence of the human race. For a long time, natural spaces and their biodiversity have been polluted, ignored and displaced, weakening the relationship between people and the environment. In the past, nature flourished throughout the university's grounds, and the *Jardín de los Sentidos* occupied a central place within that ecosystem. It was the heart of all the flora in the realm and maintained a connection with the rest of the green spaces via an underground network formed by conduits covering the plants' roots, as can be seen in Figure 4.3.

Figure 4.3: Mycorrhizae by BBC World Service[1]

These conduits were made up of mycorrhizae, the symbiotic relationship between a fungus's mycelium

[1]Resource: BBC World Service [18].

and a plant's roots, and enabled the *Jardín de los Sentidos* to send nutrients, signals and information to the rest of the flora. Thanks to this network, the plants were able to remain connected, and the garden functioned as a living organism capable of perceiving, remembering and transmitting knowledge through its senses.

At present, this network is fragmented and weakened. The garden's senses have lost part of their memory, and Deeproot can no longer transmit information correctly via the mycorrhizae. Even so, the network can still be restored. To achieve this, players must explore the *Jardín de los Sentidos*, complete the various missions associated with each sensory zone, and help restore the lost connections.

Once the group has managed to restore the garden's senses, Deeproot will once again be able to channel information through the mycorrhizal network. The restoration of this connection symbolises the recovery of the bond between humans and the natural environment, as well as the ability to observe, understand and appreciate the biodiversity that forms part of the garden.

4.1.3. GAME LOOP

The game loop in *Micorrizas* is structured around the physical exploration of the *Jardín de los Sentidos* and the triggering of quests via geolocation. As Figure 4.4 shows, the experience begins on the connection screen, where one player creates the game as the host and the other participants join using a code. Once the session has been established, all players access the map scene, which serves as the central hub for navigation and group coordination.

Upon entering the map scene, the system checks the players' positions. If the group is outside the *Jardín de los Sentidos*, Deeproot appears to indicate that assistance can only begin once everyone has arrived at the garden. This initial condition links the start of the game to physical presence in the space and reinforces the geolocation-based nature of the experience.

Once all players are within the playable area, Deeproot introduces the main objective: to explore the garden and find the missions needed to repair the mycorrhizal network. From this point onwards, the group can move freely around the map. Progression follows a non-linear structure, as the five missions associated with the senses can be completed in any order. This decision allows the journey to adapt to the group's actual movement through the garden and avoids imposing a rigid sequence on the physical space.

The main loop is repeated in each sensory zone. Players explore the environment, enter a mission's activation area, receive the corresponding narrative context, solve the mission cooperatively, and receive feedback on their result. Upon completing the mission, the system records the group's score, marks that zone as completed, and returns the group to the map to continue their exploration. The group of adventurers can only complete each mission once. However, they can replay the entire game to achieve a higher score, as if they score less than 5, they will not have succeeded in restoring the mycorrhizal network.

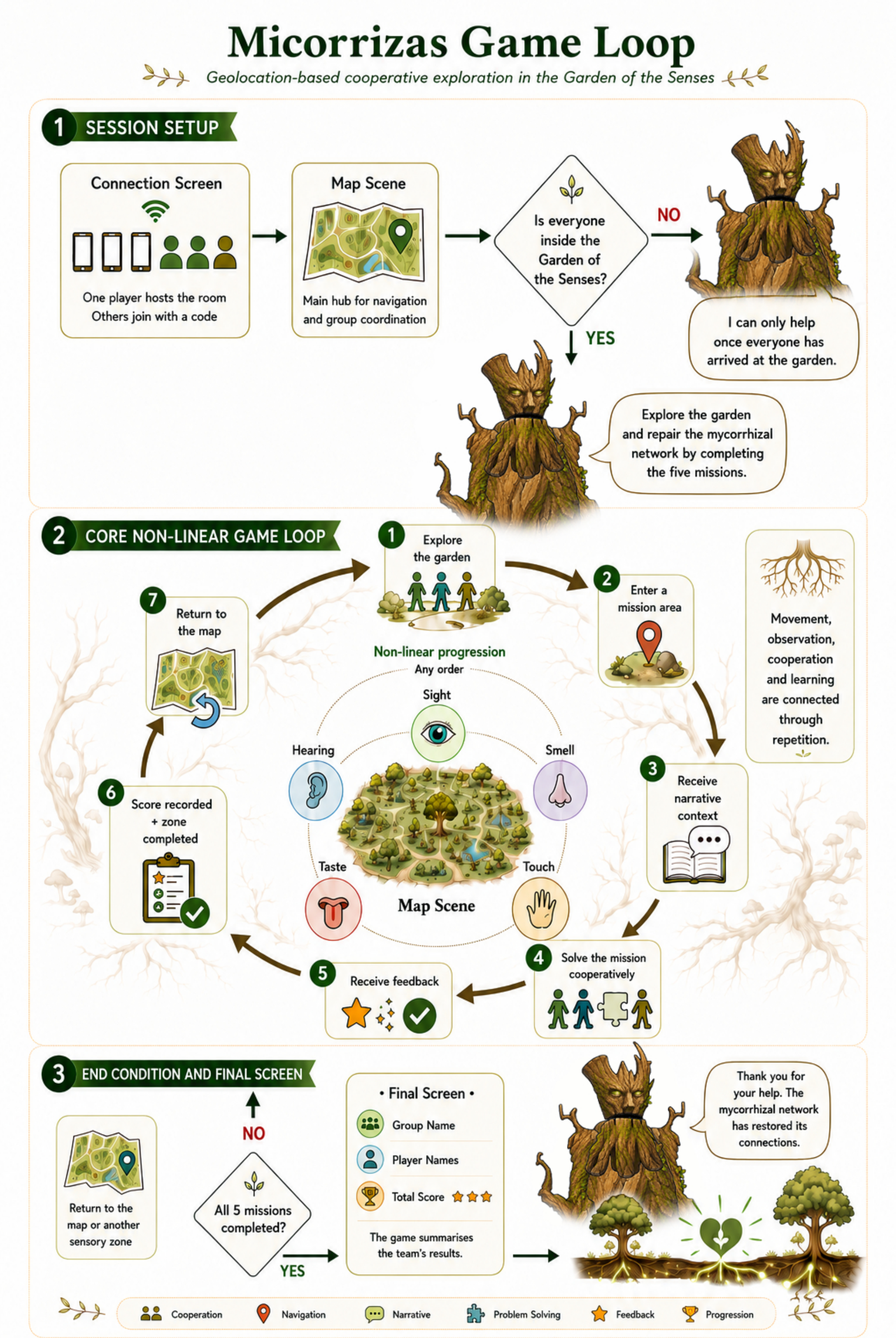

Figure 4.4: Micorrizas game loop

Once all five missions have been completed, the game proceeds to the final screen. This scene displays the group's name, the participants' names and the group score achieved during the session. Deeproot concludes the experience by thanking the players for their help and explaining that the mycorrhizal network has restored its connections. This conclusion serves a dual purpose: it summarises the team's results and provides a narrative conclusion to the journey through the garden. These results can then be used by the course lecturers to mark the activity on which the project is based.

4.1.4. SCENES FROM THE GAME

The *Micorrizas* experience is organised into several scenes, each one associated with a specific function within the overall structure of the game. These scenes separate the main stages of the experience: creating or joining a cooperative session, exploring the garden, completing the sensory missions, and presenting the final result.

Lobby system scene: This scene is used to start the game and create the cooperative session. One player acts as host and generates a game code (see Figure 4.5), while the remaining players use that code to join the same session. Its main purpose is to ensure that all participants are connected before the shared experience begins.

Figure 4.5: Lobby scene in game

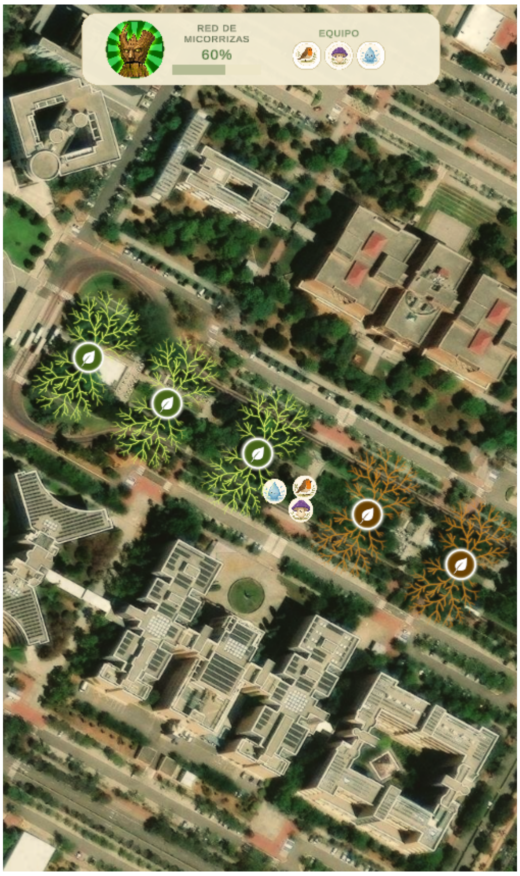

Figure 4.6: Map scene in game

Map scene: This is the main navigation screen of the game (see Figure 4.6). It represents the *Jardín de* *los Sentidos* using ArcGIS and displays the position of the players within the physical space. From this scene, the system detects the sensory zones, activates the corresponding missions and shows the group which areas have already been completed.

Quest scenes: These scenes contain the main playable activities. They share a common interface structure, with a header that presents the mission objective, the relevant score or progress, and the instructions needed to complete the activity. Each mission is completed cooperatively, and the result obtained is shared by all members of the same session. At the end of each mission, the group receives a score between 0 and 10, which contributes to the overall progress of the game.

Garden of Sight mission: In this scene, plant species images and the corresponding scientific names are shown using cards.

Garden of Hearing mission: The scene presents an interface centred on an audio player together with a distorted black-and-white image that provides a visual clue related to the sound being played.

Garden of Touch mission: The scene displays a sequence of textual clues describing the physical and sensory characteristics of a plant, together with the interface elements required to submit the group's answer.

Garden of Taste mission: The scene contains a set of face-down cards arranged in a grid, displaying illustrations of plants and fruits that are revealed as the activity progresses.

Garden of Smell mission: The scene presents a collection of plant illustrations together with several labelled category areas, allowing players to organise the elements within the interface.

Final result scene: This scene presents the conclusion of the experience after the five main missions have been completed. It displays the group name, the names of the participants and the final score achieved. Deeproot also appears again to close the narrative and thank the players for helping to restore the mycorrhizal network.

4.1.5. GAME MECHANICS

The game mechanics of *Micorrizas* are organised into two levels: a core mechanic common to the entire experience, and a set of specific game mechanics associated with each mission.

The core game mechanic is the physical exploration of the *Jardín de los Sentidos*. Players advance by moving through the real space, and the system uses geolocation to detect when the group enters a sensory zone. When everyone has been in that area for 5 seconds, the corresponding mission is triggered and the player's action shifts from the map to the quest scene. Figure 4.7 shows the area corresponding to each zone.

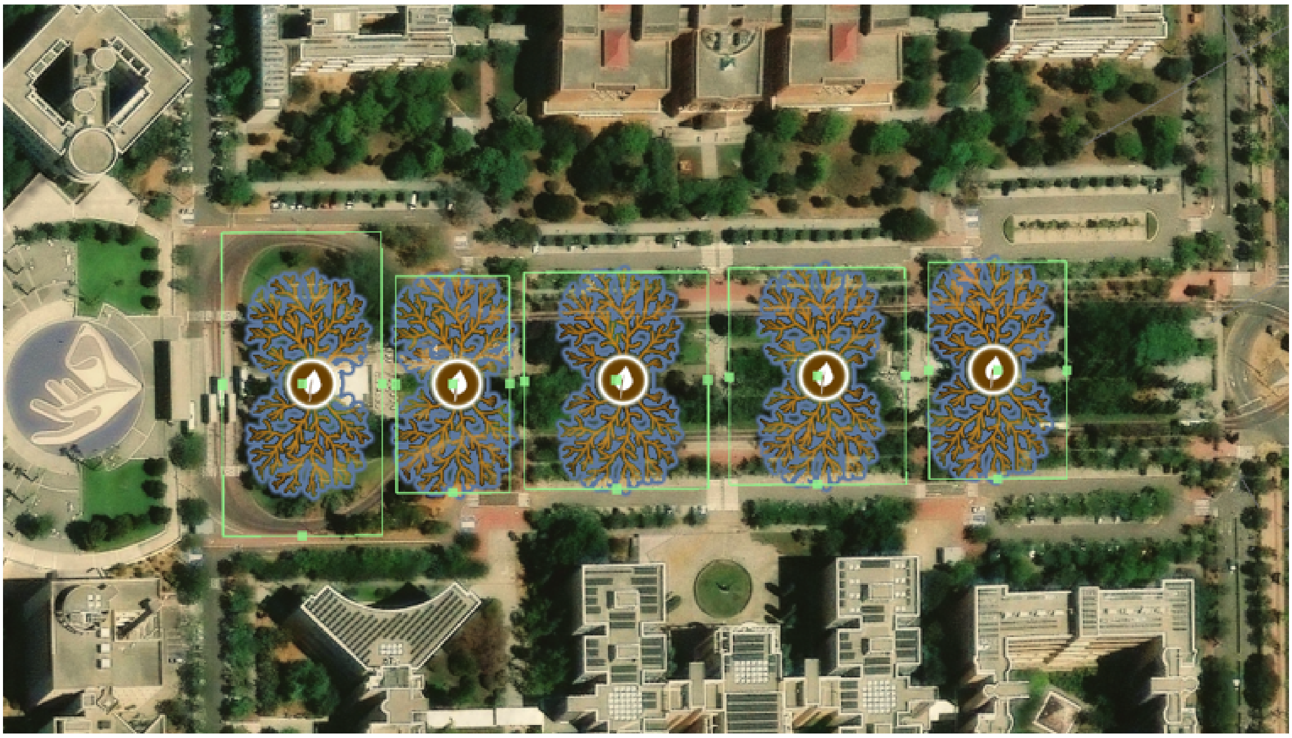

Figure 4.7: Triggers of the garden

This main game mechanic allows movement through the garden to form part of the gameplay itself. Players use the map to orient themselves, locate pending zones and check the group's progress. Geolocation links the participants' real-world position to the narrative and gameplay progression, so that each mission is tied to a specific location within the environment.

The specific game mechanics associated with each mission are described below.

Garden of Sight

In the Garden of Sight, the mission uses a card-swiping mechanic (see Figure 4.8). In each round, a card appears showing an image of a plant and its scientific name. Players must decide whether that species has been observed in the Garden of Sight. To answer, they swipe the card to the right if they believe it belongs in that area, and to the left if they believe it does not. This interaction is well suited to mobile devices because it is quick, straightforward and easy to understand. The mission develops visual discrimination, as players must compare the information on the card with what they are observing in the real environment.

Figure 4.8: Quest in the Garden of Sight with card-swiping mechanic

Garden of Hearing

In the Garden of Hearing, the mission combines sound identification and cross-device cooperation. One player plays an audio clip related to the garden and the other devices display possible answers (see Figure 4.9 and 4.10). The correct answer appears on only one of the mobiles, so the group must communicate to locate and select it. The mechanics draw on exercises involving the association of sound and word, alongside a distribution of answers among participants to reinforce collaboration. After each round, the device responsible for the audio changes, allowing different players to take on different roles during the mission.

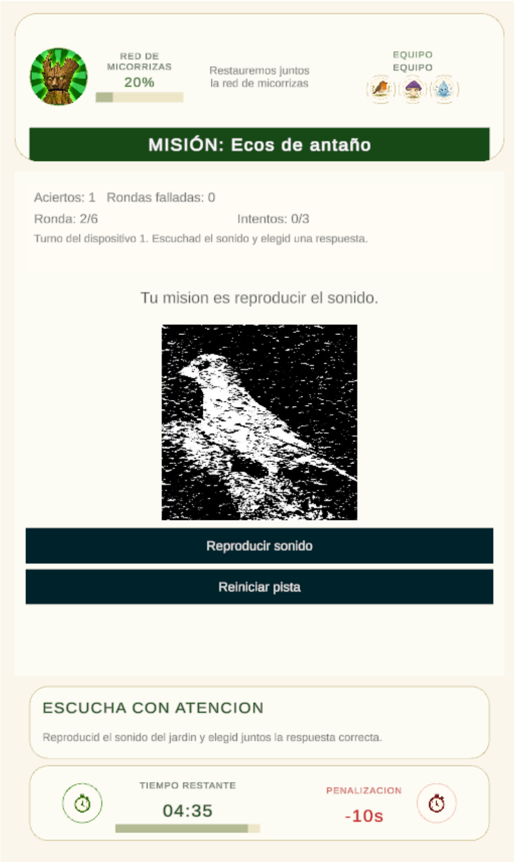

Figure 4.9: Point of view of the player playing the audio

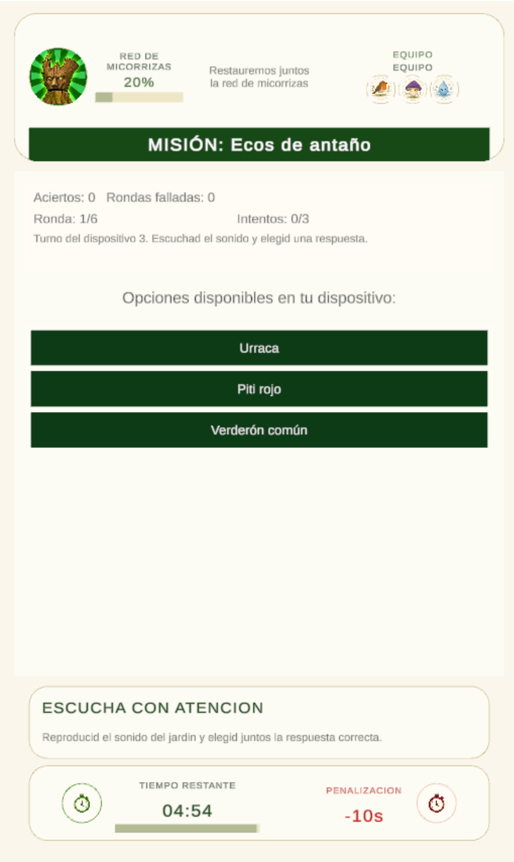

Figure 4.10: Point of view of the player with answers

Garden of Touch

In the Garden of Touch, the mission is based on a progressive deduction mechanic. Players must write the name of a plant that might be found in that area of the garden. Each attempt reveals certain information. Initially, the system reveals general characteristics, such as roughness or shape. As new attempts are made, more specific clues are unlocked, such as characteristics of the fruit or type of plant. Furthermore, all these characteristics have a background colour: green if that information matches the plant in question, orange if it is similar, and red if it is completely unrelated (see Figure 4.11).

Figure 4.11: Quest in the Garden of Touch, trying to guess what kind of plant it is

Garden of Taste

In the Garden of Taste, the mission uses a memory-matching mechanic (Figure 4.12 and 4.13). Each device displays several face-down cards relating to fruits or plants in the garden. Players can reveal cards on their own devices, but pairs are never complete on a single mobile. To solve the mission, the group must communicate which cards each participant has discovered and coordinate to find the correct matches.

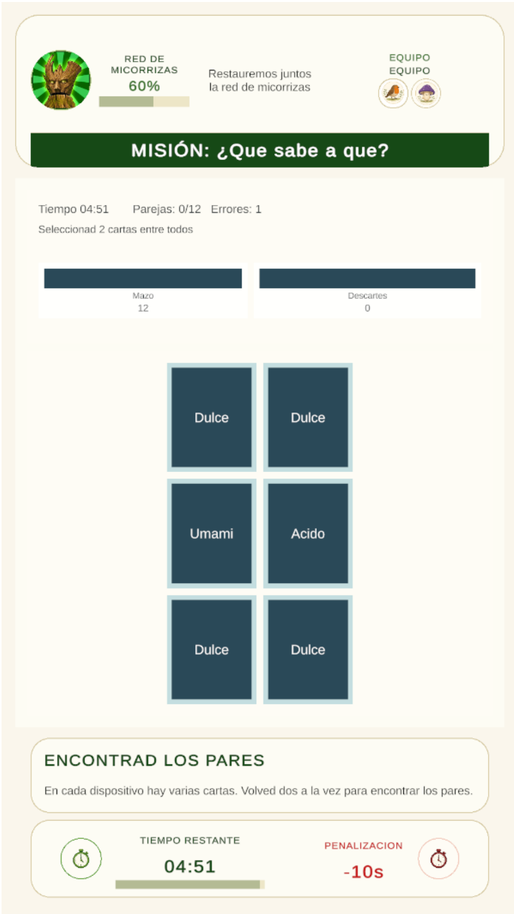

Figure 4.12: Quest in the Garden of Taste with cards face down

Figure 4.13: Quest in the Garden of Taste unsing a memory-matching mechanic

Garden of Smell

In the Garden of Smell, the mission uses a drag-and-drop mechanic. Players are given cards bearing their own image and name, and must sort them into categories based on their use or function, such as decorative, culinary or medicinal. The interaction involves dragging each card to the corresponding section.

All quests share certain common elements. Firstly, there is cooperation, as challenges are solved as a group; this leads to shared progress, which tracks the areas completed and, in turn, results in a group score that applies to all players in the same room. Then there is the interface and the feedback system. The interface features the same elements: a top bar and a bottom bar. The feedback system uses a shaking animation and colours; as well as deducting points from our group of adventurers, it also subtracts 10 seconds from the time they have to complete each quest. These elements ensure that the game maintains its consistency, even though each quest uses a different form of interaction.

4.1.6. DESIGN DECISIONS DUE TO PEDAGOGICAL OBJECTIVES

The design of each quest was defined by combining the educational purpose of the original activity with interaction models suitable for mobile devices. Although each quest focuses on a different sensory experience, all of them were designed to encourage observation, discussion and collaboration, maintaining simple interactions appropriate for any user, regardless of their level of familiarity with technology.

The Garden of Sight quest is based on a swipe interaction. When a plant card appears on screen, players slide it to the right if they consider that the species has been observed in the Sight area, and to the left if they consider that it does not belong to that area. After each decision, the system provides feedback and updates the group score.

This interaction is simple, fast and well suited to mobile play. The binary gesture reduces interface complexity and allows players to focus on the recognition task. Since only one stimulus is presented at a time, the player's attention remains centred on the plant profile and on the comparison with the real garden.

From a pedagogical perspective, the quest reinforces visual discrimination by requiring players to compare visual evidence, recall their observations during the route and make informed decisions. Immediate feedback helps them identify mistakes and progressively improve their understanding of the plants found in the garden.

Unlike the Sight quest, which relies primarily on visual recognition, the Sound mission focuses on auditory perception and communication between players. Its structure was inspired by two types of activities: audio-based association exercises, such as Duolingo [19] (see Figure 4.14), and collaborative answer distribution, similar to the dynamic used in Quizlet Live [20] (see Figure 4.15 and 4.16). In each round, one device plays an audio clue related to a sound found in the garden, while the rest of the devices display different possible answers.

Only one of the devices contains the correct answer, requiring players to communicate, compare the available options and decide collectively which response should be selected. Once the answer has been submitted, the roles rotate so that every participant alternates between listening to the audio clue and providing possible answers. This design distributes responsibility across the group and encourages active participation throughout the activity. Educationally, the mission develops sound recognition and association, as well as strengthening communication and collaborative decision-making.

If you fail to choose an answer, the option will be disabled, the distorted image will be revealed a little more, as shown in these Figures 4.17, 4.18, 4.19 and 4.20, and 10 seconds will be deducted from the time you have to complete that quest.

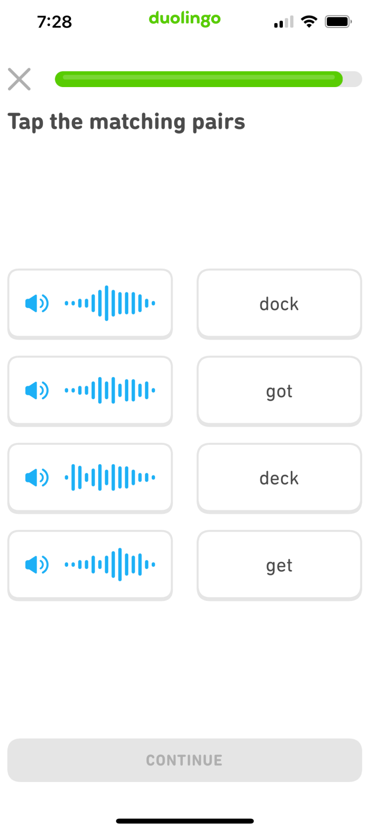

Figure 4.14: An example of how Duolingo works

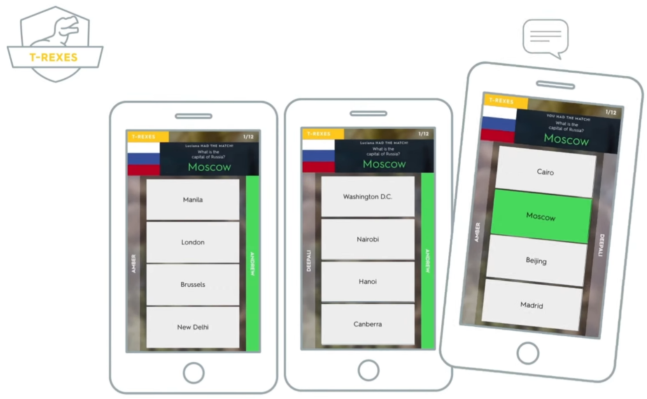

Figure 4.15: An example of how Quizlet Live works

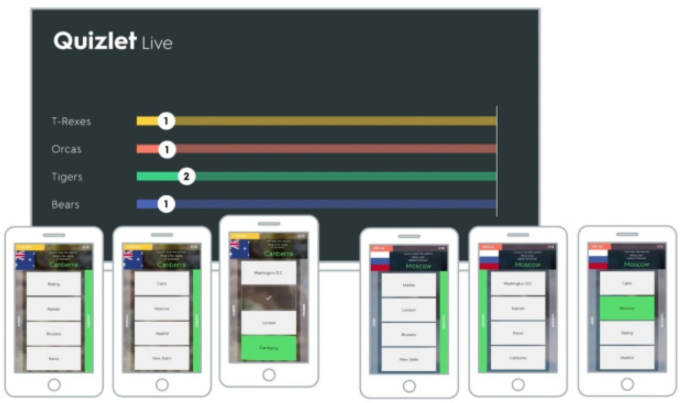

Figure 4.16: Another example of how Quizlet Live works

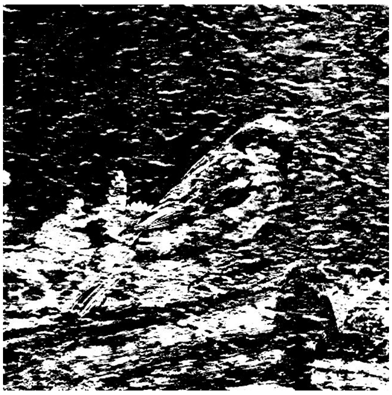

Figure 4.17: The clue for the quest in the Garden of Hearing: opening image

Figure 4.18: The clue for the quest in the Garden of Hearing: next step a clearer image

Figure 4.19: The clue for the quest in the Garden of Hearing: next step, the image with the minimum filter

Figure 4.20: The image shown as a solution

The mission of the Garden of Touch follows a different approach by presenting players with progressively more specific clues describing the physical and sensory characteristics of a plant. Rather than relying on immediate recognition, the mechanic encourages participants to discuss possible alternatives, compare their observations and gradually eliminate incorrect options before reaching a collective decision. This progression promotes critical thinking and maintains the interaction straightforward and easy to understand.

Similarly, the mission of the Garden of Taste adapts the traditional memory-card game into a cooperative experience. Instead of each player having access to all the information, cards are distributed across multiple devices, requiring participants to communicate the positions and contents they have discovered in order to complete the pairs successfully. As a result, a familiar mechanic becomes an activity that promotes teamwork, shared memory and information exchange.

Finally, the mission of the Garden of Smell requires players to classify different plant species according to categories such as decorative, culinary or medicinal uses. The drag-and-drop interaction provides an intuitive way of organising the information and keeps the information and keeping the interface clear an easy to navigate. From an educational perspective, this mechanic reinforces the relationship between plant species and their practical applications, encouraging players to discuss and justify each classification before reaching a common answer.

4.1.7. GAME ART

The visual design of *Micorrizas* was created to support the educational and environmental tone of the project. Since the game takes place in the *Jardín de los Sentidos* and focuses on biodiversity, the art direction follows a natural and friendly aesthetic. The predominant colour is green, which evokes vegetation, nature and ecological restoration. This is complemented by brown and earthy tones, especially in elements related to Deeproot, trees, roots and the mycorrhizal network.

The game combines two visual approaches. The map has a realistic function, as it represents the physical space where the players move. Its purpose is to help players understand their position in the garden and connect the digital experience with the real environment. In contrast, the character, interface elements and icons use a simpler and cleaner style, based on flat colours, clear shapes and readable visual elements. This contrast allows the map to preserve its connection with the real location and keeps the interface accessible on a mobile screen.

The interface was designed to be simple and direct because the game is intended to be played outdoors. Players need to understand the information quickly as they move through the garden and interact with the rest of the group.

Deeproot is the most important character in the visual identity of the game. His design had to communicate his connection with nature, age and wisdom, and also make him approachable as a guide for the players. Figure 4.21 shows the Ent's silhouette created in Illustrator as a vector drawing, so that it can subsequently be animated in Unity in stages. The final texturing was then developed with the help of AI to achieve a bark-like appearance, with moss and natural details (see Figure 4.22). This process resulted in a character that fits the game's fantasy narrative and maintains a visual connection with the garden and its vegetation.

The missions also required specific visual resources. Some assets were created or adapted for the project, and others were selected from external asset libraries such as the Unity Asset Store and Itch.io, like the dialogue box or the background texture (see Figure 4.23 and 4.24). In the Garden of Hearing mission, the images used as visual clues were processed with filters so that they did not reveal the answer too directly.

Audio resources were also part of the artistic and sensory design of the game. Since the Garden of Hearing mission is based on recognising sounds related to the natural environment, sound selection was especially important. For this purpose, specialised sound databases such as Xeno-canto [10] were used to search for accurate recordings. The selected sounds were recorded in Spain, which helped to keep the audio material coherent with the geographical and environmental context of the project.

In the Testing and Validation section 5.6, it will be noted that a way was needed to identify the areas of the garden. Thanks to Águeda, this pair of assets were designed to emulate mycorrhizae: the brown

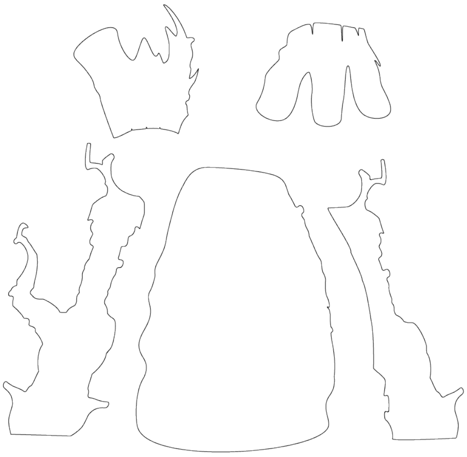

Figure 4.21: Line art

Figure 4.22: AI Texturing

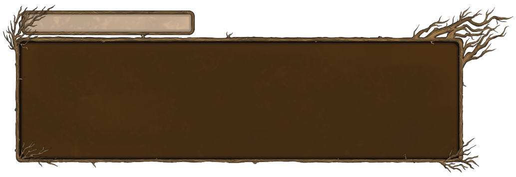

Figure 4.23: Dialogue box

Figure 4.24: UI board

one (Figure 4.26) for when the network is still damaged, and the green one (Figure 4.27) for when the adventurers have repaired the network. The *Micorrizas* logo shown in Figure 4.25 was designed by Luis to reinforce the visual identity of the game and to make the project easily recognisable.

Figure 4.25: The *Micorrizas* logo

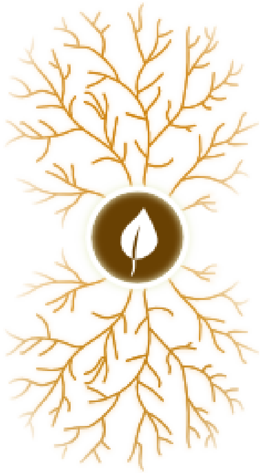

Figure 4.26: Mycorrhizal network icon

Figure 4.27: Repaired network area icon

4.2. TECNICAL ARCHITECTURE

The project's architecture is organised into four main layers: connection and cooperative session, geolocation, spatial representation in ArcGIS, and the cooperative mission layer. Each block has a specific responsibility and communicates with the others via events, synchronised state structures and scene transitions, as can be seen in Figure 4.28. Appendix 8.3 provides a much more detailed and technical explanation of this matter.

Firstly, there is a connection and session layer, responsible for launching the application and enabling players to create a co-operative session or join an existing one. This layer establishes the connection between devices before the game begins.

Above this lies the session layer, which maintains the game's shared global state. In other words, this layer ensures that all players are working with the same information: which mission is active, what stage the game is at, or which elements have been completed.

The geolocation layer retrieves the device's actual position via GPS and converts it into information useful for the game. In this way, the player's physical location can be synchronised with the other participants and utilised within the co-operative experience.

Next, the map and spatial representation layer transforms the geographical coordinates into visible and interactive elements within the game, such as markers, trigger zones or points of interest. Thanks to this layer, the real-world space of the *Jardín de los Sentidos* becomes part of the playable world.

Finally, the mission layer organises the game's activities. This layer is built on a common foundation that allows the general flow of missions to be reused, their results to be managed, and the transition between their different phases to be controlled.

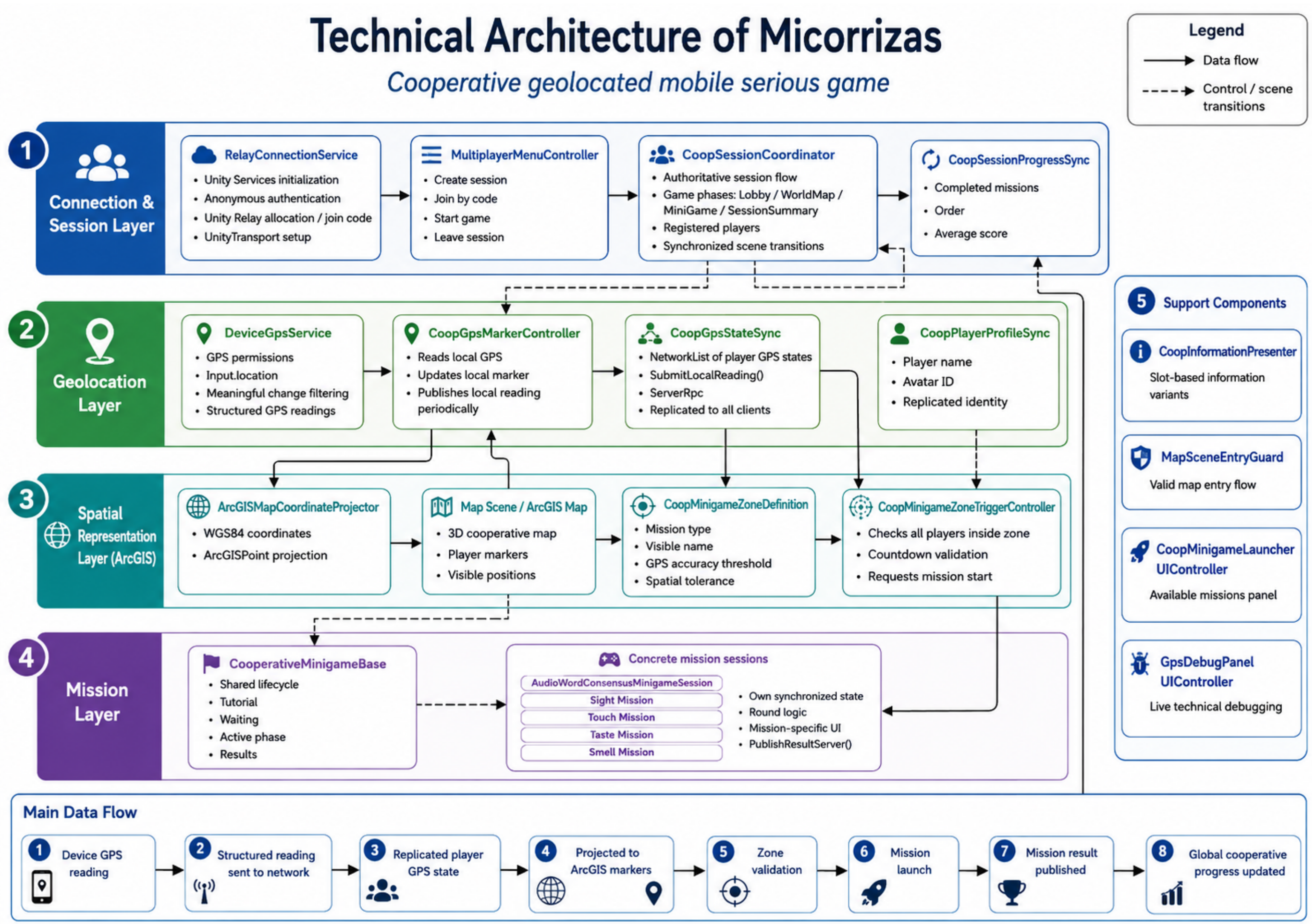

Figure 4.28: Micorrizas Architecture

On a practical level, this architecture relies on tools such as Unity Relay, to facilitate connections between players; Netcode for GameObjects (they are Unity entities, which you can copy and modify), to synchronise information between devices; and the ArcGIS Maps SDK, to work with maps and geographical coordinates. In addition, proprietary services and drivers have been developed to clearly separate data collection, synchronisation, visual representation and the user interface.

The overall architecture begins with the connection and session layer (Point 1 in Figure 4.28), where the game initialises Unity Services, creates a co-op session via Unity Relay, or allows players to join a game using a code. Once the session has been created, the system maintains the game's global state, the game stages, the connected players and shared progress. Next comes the geolocation layer (Point 2 in Figure 4.28), which retrieves the GPS position of each mobile device. This position is filtered, structured and sent to the network so that all players can view the group's location status in a synchronised manner.

The spatial layer (Point 3 in Figure 4.28), using ArcGIS, transforms these real-world coordinates into positions within the game map. Thanks to this layer, the Garden of the Senses is represented by markers, mission zones and activation areas that can be used within the experience.

When all players are within a valid zone, the system launches the corresponding quest. The mission layer (Point 4 in Figure 4.28) manages the common workflow for each activity: tutorial, waiting period, active phase and results. Each specific quest reuses this framework, whilst maintaining its own logic and interface.

Finally, upon completing a mission, the result is published in the co-op session (Point 1 in Figure 4.28) and the group's overall progress is updated. Thus, the entire workflow ranges from the device's GPS reading to spatial validation, the triggering of the quest and the synchronisation of the shared result.

4.3. PROBLEMS AND SOLUTIONS

During development, the main difficulties were related to geolocation, mobile testing and the integration of several technical systems into a single gameplay flow.

One technical issue appeared at the beginning of the project because the initial ArcGIS SDK project was created using Unity HDRP. This pipeline was not suitable for the target platform, as HDRP is not compatible with Android builds in the same way as Unity URP. For this reason, the project had to be migrated from HDRP to URP before continuing development. This migration was necessary to make the application compatible with Android devices and to ensure that the final prototype could be tested in the intended mobile environment.

The first major challenge was the implementation and testing of geolocation. As the game relies on the player's position in a real-world outdoor environment, the system could not be fully validated within the Unity editor. The location service had to be tested on a real Android device, which slowed down the process and made it less predictable. To address this issue, the geolocation system was developed incrementally: first by loading and centring the map, then by limiting the rendered area, and finally by adding trigger zones for each sensory garden. The trigger zones were designed with generous radii to compensate for GPS inaccuracies.

The multiplayer system also required several design and technical decisions. The project required cross-device cooperation, but the gameplay did not require real-time character movement or complex online interaction. For this reason, the multiplayer implementation was limited to session creation, access codes, player connection, synchronisation of shared states, and group scoring. This reduced the system's complexity whilst preserving the game's cooperative structure.

Unity Relay also presented connectivity problems when tested on the university Wi-Fi network. The exact cause could not be identified, although it may have been related to network restrictions or blocked ports. For this reason, multiplayer validation was carried out using alternative connections, such as mobile data or other Wi-Fi networks.

A final challenge was integrating the quests with the rest of the application. Each mission had its own interaction model, but they all had to share the same session structure, scoring logic and progression system, so that it could be reused. To resolve this, the quests were connected to a common flow: activation from the map, execution of the challenge, calculation of the group score and return to the main progression.

4.4. DEVIATIONS FROM THE INITIAL PROPOSAL

The initial proposal was based on creating an escape room or gymkhana inspired by the activity suggested by the Early Childhood Education teachers. This first approach aimed to transform the guided visit into a sequence of challenges distributed throughout the *Jardín de los Sentidos*, using the physical space as the basis for the player's progression.

After an initial brainstorming process with the project supervisor, the escape room structure was reconsidered. The project started to move towards a mixed reality experience in which users could learn about the biodiversity of the garden through digital information layered over the physical environment. This direction was interesting from a design perspective, since it allowed the real garden to remain central while adding interactive digital content connected to its plants, sensory areas and educational value.

However, after discussing the proposal with the Early Childhood Education teachers, the mixed reality approach was identified as difficult to apply in a realistic educational context. The main limitation was the accessibility of the required technology. A mixed reality experience would depend on specific devices that are not commonly available to students or teachers, making it harder to use the project as a practical tool during the real activity.

For this reason, the project was refocused on mobile devices, whilst ensuring that the main focus of the project remained the biodiversity of the *Jardín de los Sentidos*. This decision made the game more accessible and better suited for use directly in the garden, given that most potential users already own a mobile phone. The mobile platform allowed the project to maintain a connection with the physical space and reducing the technological barriers to the experience. From that point onwards, the design focused on developing a game for Android mobiles that could combine geolocation, cooperative interaction and educational quests within the *Jardín de los Sentidos*.

CHAPTER 5
TESTING AND VALIDATION

5.1. TESTING METHODOLOGY

The project followed an iterative and incremental testing strategy. Each major feature was tested after implementation and reviewed again after integration with the rest of the system. This approach was necessary because several systems depended on each other: the map required GPS readings, mission activation used projected player positions, and each quest had to report its result to the cooperative session.

The validation process was organised into three levels:

- **Functional testing**: verification of the main application flow and each implemented subsystem.
- **Field testing**: validation of GPS behaviour, trigger zones and outdoor usability in the garden.
- **User and expert validation**: feedback on clarity, mission design, interface readability and pedagogical suitability.

The supervisor reviewed the prototype periodically. Feedback from these reviews guided the following iteration and helped identify design issues before they became structural problems. It assessed technical feasibility, usability and alignment with the original activity. The measurement of learning impact remains part of future classroom validation.

5.2. FUNCTIONAL TESTING

Functional testing verified the complete user flow: creating a session, joining from other devices, entering the map, detecting player position, launching missions, completing challenges and displaying the final result. Table 5.1 presents the functions tested and their validation objective.

The main integration issues appeared when independent systems had to share state and scene transitions. The flow between map exploration, mini-game execution and return to the session required several iterations to avoid duplicated states or inconsistent progress between clients.

Table 5.1: Main functional tests carried out during development

Tested area	Validation objective
Lobby and connection	Check that a player could create a room and the rest of the group could join using the generated code.
Session synchronisation	Verify that all devices shared the same session state, active mission and progression data.
Map scene	Check that players could access the map, see their position and understand which areas remained pending.
Mission activation	Validate that the game could detect when the group reached an activation area and launch the corresponding mission.
Mini-games	Check that each mission could be played, completed and evaluated according to its mechanic.
Scoring and final screen	Verify that mission results were accumulated and shown at the end of the session.

5.3. FIELD TESTING

Field testing focused on the behaviour that depended on real Android devices and outdoor conditions. The GPS service was the most important case, because the Unity editor could not reproduce the same readings or permissions.

The tests in the *Jardín de los Sentidos* covered four aspects:

- **GPS availability**: checking that the device obtained a valid location reading and that the application handled the waiting state.
- **Position representation**: verifying that the player marker was placed coherently after converting GPS coordinates into the ArcGIS scene.
- **Activation areas**: adjusting the size and tolerance of the mission zones.
- **Outdoor usability**: observing whether the interface remained understandable while users moved through the physical space.

GPS precision behaved as an approximate value. Small variations appeared even when the user was standing still, and accuracy changed depending on the device and conditions. The activation zones were therefore designed with generous margins, so mission access did not depend on an excessively precise coordinate.

Field testing also improved map feedback. Completed zones received a clearer visual change after their mission was solved, reinforcing the narrative idea of restoring the garden and helping players understand their progress.

5.4. USER VALIDATION

The prototype was tested with users and reviewed by lecturers involved in the original educational activity. The goal was to identify whether the interface, mission instructions and cooperative flow were understandable without external explanation.

User testing used a think-aloud method. Participants verbalised their decisions while interacting with the application, making it easier to detect confusing instructions, unclear visual elements and moments where the next action was ambiguous.

Lecturer feedback focused on the relationship between the missions, the sensory gardens and the expected learning experience. This review was especially useful for adjusting the application to the pedagogical context of the subject.

5.5. FIRST EVALUATION CYCLE

The first evaluation cycle focused on mission clarity and map behaviour. The prototype already included the main cooperative structure, although certain aspects still needed refinement. Main changes made in this evaluation cycle are detail in Table 5.2.

Table 5.2: Main changes derived from the first evaluation cycle

Detected issue	**Implemented improvement**	**Effect**
The sound-recognition mission was difficult to solve using only audio cues.	Distorted images were added. Each error is less distorted than the previous one.	The task became easier to interpret while keeping the listening component.
Players needed clearer information about mission activation areas.	The areas on the map were delineated using an image of the mycorrhizae.	Players could better understand where they needed to move.
Completed areas were visually too similar to pending areas.	Once the mission is complete, the image of the mycorrhizae in that area is updated to show that they have now regenerated.	The map communicated restored areas more clearly.
Some mission instructions required extra explanation.	Texts and interface distribution were adjusted in the affected scenes.	The interaction flow became more autonomous.

This cycle improved usability in two key points: the sound mission became easier to understand and the map communicated progression more clearly.

5.6. SECOND EVALUATION CYCLE

The second evaluation cycle incorporated feedback from the lecturers. The review refined the link between each mission and its sensory area, and improved readability on mobile screens. Table 5.3 represents the main changes made in this second evaluation cycle.

Table 5.3: Main changes derived from the second evaluation cycle

Feedback	Implemented improvement	Effect
The GPS state needed clearer feedback.	The GPS avatar indicator was adjusted.	Players could better interpret the location.
Some screens required better readability and hierarchy.	The user interface was reorganised, and the colours and shapes were changed.	Relevant information became easier to scan during play.
The plant images in the first mission needed greater visibility.	The size of the plant images was increased.	Visual recognition became easier on mobile screens.
The fourth mission needed a clearer link with the Garden of Taste.	A reference was added to the flavours of each fruit.	The activity became more directly connected to its sensory area.
The fourth mission needed more tension.	A timer was added as a stressor.	The mission became more dynamic.

This cycle strengthened the educational coherence of the missions and improved the mobile presentation of key information.

5.7. SUGGESTIONS NOT IMPLEMENTED

Some suggestions were analysed and left outside the final prototype because they increased complexity or interrupted the outdoor flow. Table 5.4 represents suggestions considered but not implemented in both evaluation cycle.

Table 5.4: Suggestions considered but not implemented

Suggestion	Reason for not implementing it
Add a separate learning scene before the sound quest.	The tutorial scene was not included because it slowed down the experience. Instead, the decision was made to add images of the species that can be heard, but to distort them using a filter so that this area would not become the visual focus
Add an additional learning phase before the fourth quest.	The teachers' proposal was adopted instead of this one (see section 5.6), because it was more closely aligned with the Garden of Taste.
Prevent a player from revealing more than one card on the same device in the fourth quest.	It makes the mission unnecessarily difficult.

5.8. LIMITATIONS OF THE VALIDATION

The number of users was reduced, so the tests are useful for detecting usability and technical problems, while they do not provide representative data about the target student population.

Classroom validation with students remains pending. It could be considered for next year, starting with a small number of pupils and then, in December when the lesson takes place, giving all the pupils the opportunity to take a real test.

No pre-test and post-test study was carried out. The validation therefore did not measure changes in biodiversity knowledge, environmental awareness or perception of the *Jardín de los Sentidos*.

The scoring system was validated as part of the game flow. Its relationship with the assessment criteria of the original activity remains open for future work.

For the scope of this Bachelor's Thesis, the validation provided enough evidence to refine the prototype, check the Android flow and integrate feedback from the educational context.

CHAPTER 6

RESULTS

The final result of the project is a functional Android application that transforms the educational activity of the *Jardín de los Sentidos* into a cooperative, geolocated serious game. The application allows players to create a multiplayer session, join it using a code and take part in a shared experience in which the group explores the real garden and completes missions linked to its sensory areas.

The application includes the main systems required to support the intended experience. It integrates geolocation so that the player's real position can be used inside the game, a map scene that acts as the central navigation space, a cooperative multiplayer structure that keeps the session synchronised between devices, and a set of missions connected to the senses represented in the garden. The game also includes feedback after each mission and a final screen showing the group name, the participants and the score achieved during the session.

From a gameplay perspective, the result is a complete playable flow, as you can see Figures 6.1, 6.2, 6.3, 6.4, 6.5 and 6.6. Players can start a session, access the map, move through the garden, activate missions, solve them cooperatively and obtain a final result. The experience is structured around exploration, observation, communication and group decision-making, maintaining the relationship with the physical environment throughout the game.

From an educational perspective, the application provides the lecturers with a digital tool that can be used to support the teaching session related to the *Jardín de los Sentidos*. The final score and the group result can be used as part of the activity carried out in class, allowing the teachers to assess participation and performance during the session. In this sense, the project has produced a usable prototype that can be incorporated into the educational context for which it was designed.

Figure 6.1: Quest in the Garden of Sight area

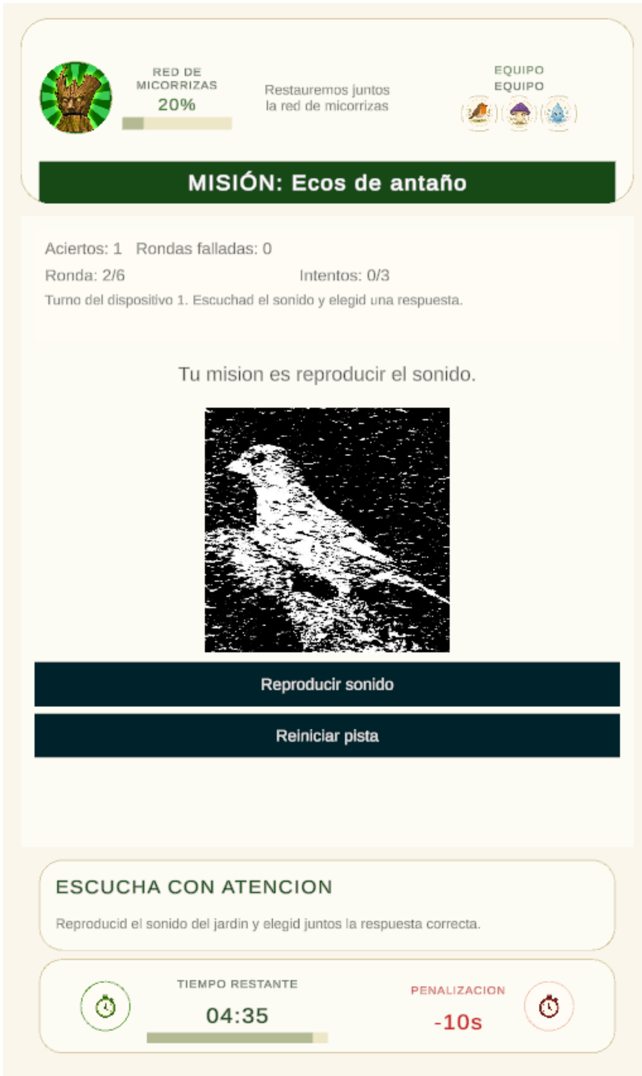

Figure 6.2: Quest in the Garden of Hearing area

Figure 6.3: Quest in the Garden of Touch area

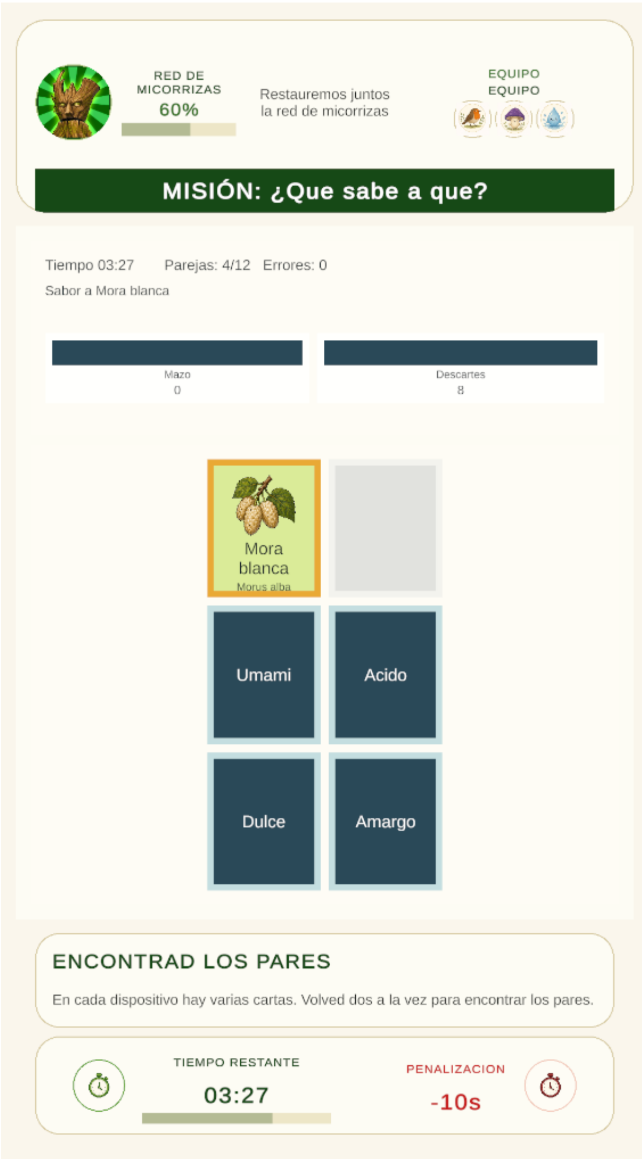

Figure 6.4: Quest in the Garden of Taste area

Figure 6.5: The dialogues with Deeproot, the companion and narrative guide in this story

Figure 6.6: End of the game: display the team's score and its members

CHAPTER 7

CONCLUSIONS AND FUTURE WORK

7.1. CONCLUSIONS

The development of *Micorrizas* has made it possible to create a functional Android serious game based on a real educational activity. The project has achieved its main technical objective by producing a mobile application that combines geolocation, cooperative multiplayer interaction, mission-based gameplay and a final group result. These elements work together to support an outdoor learning experience in the *Jardín de los Sentidos*.

One of the main conclusions of the project is that the game is not simply a gamified version of an existing activity. During development, it became clear that the original visit was being used as the basis for designing a serious game with its own structure, narrative, mechanics and progression. The educational activity provided the context, contents and objectives, but the final result required specific design decisions in order to transform that material into an interactive experience.

The project has also shown the value of connecting mobile gameplay with a real environment. By using the garden as the playable space, the game encourages players to observe, move, communicate and make decisions in relation to the physical surroundings. This supports the objective of promoting awareness and appreciation of biodiversity through interaction with the natural environment.

However, the educational impact of this objective cannot yet be fully measured. The application is prepared to be introduced as a tool in the subject, and the lecturers have expressed the intention of using it in a real teaching session during the next academic year. Once the game is used with students in that context, it will be possible to evaluate more precisely whether the experience increases their awareness, appreciation and understanding of the biodiversity of the *Jardín de los Sentidos*.

Working on mycorrhizae has enabled me to put much of the knowledge I acquired during my degree into practice, as well as to incorporate new knowledge that has proved necessary during the course of the project.

Overall, the result achieved is satisfactory, as the game meets the project's expectations and demonstrates the feasibility of building a geolocation-based cooperative experience in Unity.

7.2. FUTURE WORK

The main line of future work is the use of the application in a real session of the subject. This would allow the game to be tested with its intended users and under authentic teaching conditions. The results of that implementation could be used to evaluate the educational impact of the game, especially in relation to student engagement and collaborative learning.

A future evaluation could combine several methods, such as observation during the activity, questionnaires before and after the session, teacher feedback and analysis of the final group scores. This would make it possible to assess not only whether the application works technically, but also whether it contributes to the learning objectives of the original activity.

From a technical perspective, future work could focus on improving GPS accuracy and robustness during outdoor use. Although the prototype already includes geolocation-based mission activation, further testing with larger groups and different mobile devices would help to refine activation areas, GPS indicators and tolerance margins.

The content of the game could also be expanded. New plant species, sounds, images and questions could be added to enrich the missions and make the experience more varied. Additional missions could also be designed for other botanical or educational spaces, allowing the structure of *Micorrizas* to be reused beyond the *Jardín de los Sentidos*.

Finally, the teacher-facing part of the application could be improved. Future versions could include tools to export results, review group performance or configure the content of the missions before the session. These improvements would make the application more useful as a teaching resource and would support its integration into the subject in a more stable way.

7.3. PROJECT APPLICATIONS

Micorrizas has been developed as a tool that can be applied within the subject related to the natural environment in Early Childhood Education. Its main application is to complement the existing visit to the *Jardín de los Sentidos* by providing a structured, interactive and cooperative activity that students can complete in the real space. If any problems arise, such as poor lighting because the activity is taking place late in the day or adverse weather conditions.

The game can be used by lecturers to guide students through the garden without requiring constant direct instruction. The application gives players objectives, activates quests according to their position and provides feedback on their answers. This allows the teaching activity to become more autonomous while preserving the importance of direct contact with the natural environment.

Another possible application is its use as an assessable class activity. Since the game records the group's progression and provides a final score, the lecturers can use the result as part of the evaluation of the session. This does not replace the educational value of observation and discussion, but it provides an additional element that can help structure and assess the activity.

The project also creates a foundation for future educational uses. New plant species, sounds, questions, clues or missions could be added to adapt the experience to future editions of the subject or to different learning objectives. Therefore, the application is not limited to the current application, but can be expanded as a reusable educational resource.

BIBLIOGRAPHY

[1] UNESCO. *Green School Quality Standard: Greening Every Learning Environment*. UNESCO, 2024. URL: https://www.unesco.org/sites/default/files/medias/fichiers/2024/09/Green%20school%20quality%20standard%20Greening%20every%20learning%20environment.pdf (visited on 06/22/2026).

[2] Claudio D Rosa, Christiana Cabicieri Profice, and Silvia Collado. "Nature experiences and adults' self-reported pro-environmental behaviors: The role of connectedness to nature and childhood nature experiences". In: *Frontiers in psychology* 9 (2018), p. 1055. DOI: https://doi.org/10.3389/fpsyg.2018.01055.

[3] Caroline Hands. "Serious Video Games: Tools for Learning, Training and Health". In: *Encyclopedia* 6.4 (2026), p. 83. DOI: https://doi.org/10.3390/encyclopedia6040083.

[4] Antonio Pio Facchino et al. "The use of serious games for psychological education and training: a systematic review". In: *Frontiers in Education* Volume 10 - 2025 (2025). ISSN: 2504-284X. DOI: 10.3389/feduc.2025.1511729. URL: https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1511729.

[5] Á Gómez-Cambronero et al. "Mixed reality escape room video game to promote learning in indoor environments". In: *Edulearn19 Proceedings*. IATED. 2019, pp. 5857–5864.

[6] Alessandro De Gloria, Francesco Bellotti, and Riccardo Berta. "Serious Games for education and training". In: *International Journal of Serious Games* 1.1 (2014). DOI: https://doi.org/10.17083/ijsg.v1i1.11.

[7] Trello. *Trello: Organiza cualquier cosa con cualquiera*. URL: https://trello.com/es (visited on 06/19/2026).

[8] Unity Technologies. *2D Assets*. URL: https://assetstore.unity.com/?category=2d&price=0-0&orderBy=1&rows=96 (visited on 06/19/2026).

[9] itch.io. *Top free 2D game assets*. URL: https://itch.io/game-assets/free/tag-2d (visited on 06/19/2026).

[10] Xeno-canto Foundation. *Xeno-canto: Sharing Bird Sounds from Around the World*. URL: https://xeno-canto.org/ (visited on 06/19/2026).

[11] Ministerio de Trabajo y Economía Social. *Resolución de 15 de abril de 2025, de la Dirección General de Trabajo, por la que se registra y publica el XX Convenio colectivo del sector de empresas de consultoría, tecnologías de la información y estudios de mercado y de la opinión pública*. Boletín Oficial del Estado. Apr. 16, 2025. URL: https://www.boe.es/boe/dias/2025/04/16/pdfs/BOE-A-2025-7766.pdf (visited on 06/19/2026).

[12] HP. *HP OMEN 16-ap0000ns - Ordenador portátil*. URL: https://www.amazon.es/HP-OMEN-16-ap0000ns-Ordenador-port%C3%A1til/dp/B0F2TGCBXR (visited on 06/19/2026).

[13] Xiaomi. *Redmi Note 13 Pro 5G*. URL: https://www.mi.com/es/product/redmi-note-13-pro-5g/ (visited on 06/19/2026).

[14] Rentik. *Rentik - Renting de móviles*. URL: https://www.rentik.com/es/ (visited on 06/19/2026).

[15] Unity Technologies. *Cancellation of the Runtime Fee and Pricing Changes*. URL: https://support.unity.com/hc/en-us/articles/30322080156692-Cancellation-of-the-Runtime-Fee-and-Pricing-Changes (visited on 06/19/2026).

[16] Esri. *ArcGIS for Student Use*. URL: https://www.esri.es/es-es/store/products/buy/arcgis-for-student-use (visited on 06/19/2026).

[17] OpenAI. *Pricing*. URL: https://developers.openai.com/codex/pricing (visited on 06/19/2026).

[18] BBC World Service. *How trees secretly talk to each other - BBC World Service*. URL: https://youtu.be/DUqEB_tGHtw?si=Y5KUYdkmpH3qdhIm (visited on 06/22/2026).

[19] Duolingo. *Duolingo: La forma más popular del mundo para aprender*. URL: https://es.duolingo.com/ (visited on 06/19/2026).

[20] Quizlet. *How to play Quizlet Live*. URL: https://youtu.be/q64qTBfK0iE (visited on 06/19/2026).

CHAPTER 8

APPENDIX

8.1. SOURCE CODE

The complete source code of *Micorrizas* is available in a GitHub repository. This repository includes the Unity project files, scripts, scenes and resources used during development, allowing the implementation of the prototype to be reviewed.

Repository: https://github.com/Salix8/Micorrizas_JardinSentidos_TFG_Videojuegos.git

8.2. MOBILE DEVICE SIMULATION

This appendix addresses the limitations found when attempting to simulate different mobile devices using a computer. During development, an attempt was made to test the geolocation system from a desktop environment, but this was not fully possible because computers do not usually provide the same sensors or location services as mobile devices.

On mobile platforms, there is a real location service available through the operating system and the device hardware. Unity provides access to this service through its location API. The LocationService.Start() function starts location updates, and the current position can then be accessed through Input.location.lastData. In Unity's documentation, Input.location is described as a property used to access the location of handheld devices, that is, mobile devices.

In addition, on Android, when an application uses LocationService, Unity adds the ACCESS_FINE_LOCATION permission and requests location access from the user the first time the service is used. This behaviour is specific to mobile platforms.

On a PC, this process is different. A computer normally does not have GPS hardware or the same permission system used by Android or iOS. Although a PC may provide an approximate location based on IP address, Wi-Fi networks or operating system settings, this is not equivalent to reliable mobile geolocation. For this reason, the geolocation features of the project had to be validated using real mobile devices rather than relying only on computer-based simulation.

8.3. DETAILED TECHNICAL ARCHITECTURE

The project's architecture has been designed in a modular way, separating multiplayer connectivity, co-op session management, geolocation acquisition, spatial representation on the map and mission-specific logic. This separation allows each subsystem to maintain a specific responsibility, with communication between them taking place via state structures.

The entry point for the co-operative system is the RelayConnectionService.cs connection service (line 15). This component is responsible for initialising Unity's services, authenticating the user anonymously and configuring the network transport via Unity Relay. When a player creates a session, the service requests an allocation from Relay, obtains a join code and configures UnityTransport with the necessary data so that NetworkManager can start in host mode (RelayConnectionService.cs, line 121). When a player joins, the process is similar, but uses the received code to retrieve the remote allocation and start the client (RelayConnectionService.cs, line 172). In this way, Relay handles connectivity between devices, whilst Netcode for GameObjects takes over management of the network session and state replication.

Built on top of this infrastructure is the session's main coordinator, 'CoopSessionCoordinator.cs' (line 9), which acts as the authoritative hub of the co-operative flow. This component inherits from 'NetworkBehaviour' and keeps various global session elements synchronised: the current game phase via a 'NetworkVariable<CoopGamePhase>', the active mission, and the list of registered players via a 'NetworkList<ulong>' in 'CoopSessionCoordinator.cs' (line 26). The coordinator is responsible for determining whether the group is in the Lobby, WorldMap, MiniGame or SessionSummary, as well as for orchestrating synchronised scene transitions for all clients via NetworkManager.SceneManager.LoadScene(...) in CoopSessionCoordinator.cs (line 387). Furthermore, it rebuilds player slots when connections are established or lost, so that other systems can determine which index each participant occupies within the session (CoopSessionCoordinator.cs, line 356).

The lobby interface does not contain any direct network logic, but acts as a presentation layer on top of these services. In MultiplayerMenuController.cs (line 7), the buttons to create a session, join by code, start a game or leave a session invoke operations from the RelayConnectionService and the CoopSessionCoordinator. This separation allows the UI to simply reflect the session's state without duplicating business logic.

The geolocation subsystem begins in DeviceGpsService.cs (line 9), which encapsulates access to the device's GPS via Input.location. This component requests permissions, checks whether the operating system's location service is enabled, starts the sensor with configurable accuracy and distance, and manages timeouts and retries during initialisation (DeviceGpsService.cs, line 85). Each reading is converted into an immutable DeviceGpsReading object, which contains latitude, longitude, altitude, horizontal accuracy, service status, fix availability and timestamps (DeviceGpsService.cs, line 275). The service does not publish readings continuously and uncontrollably, but only when it detects a significant change via 'HasMeaningfulChange(...)', thereby reducing noise and unnecessary traffic (DeviceGpsService.cs, line 223).

The interaction between the local GPS and the cooperative system takes place via CoopGpsMarkerController.cs (line 10). This controller subscribes to the ReadingUpdated event of the DeviceGpsService, stores the latest local reading and immediately updates the player's own marker on the map in CoopGpsMarkerController.cs (line 65). Furthermore, at regular intervals, it publishes this reading over the network via gpsStateSync.SubmitLocalReading(...) in CoopGpsMarkerController.cs (line 120). This architectural decision is important because it decouples the capture of the device's position from its distribution in the multiplayer environment: DeviceGpsService merely measures, whilst CoopGpsMarkerController decides when that measurement is used for the co-operative environment.

Position synchronisation between players is handled in CoopGpsStateSync.cs (line 7). This component, which is also derived from NetworkBehaviour, maintains a NetworkList<CoopPlayerGpsState> containing the GPS status of all participants (CoopGpsStateSync.cs, line 9). When a client wishes to share its position, it calls SubmitLocalReading(...), which packages the data from DeviceGpsReading into a CoopPlayerGpsState and executes a ServerRpc to the host (CoopGpsStateSync.cs, line 49). On the server, SubmitGpsStateServerRpc(...) identifies the sender by SenderClientId and updates or inserts their entry into the shared list in CoopGpsStateSync.cs (line 94). From that point onwards, NGO automatically replicates the list to the other clients. Therefore, the actual flow is: local GPS □ structured reading □ periodic publication □ ServerRpc □ NetworkList update □ replication to the other players.

These coordinates are represented spatially using ArcGIS. The ArcGISMapCoordinateProjector.cs component (line 6) receives an ArcGISLocationComponent and assigns it an ArcGISPoint in the WGS84 system using longitude, latitude and altitude (ArcGISMapCoordinateProjector.cs, line 16). CoopGpsMarkerController uses this projector in UpdateMarker(...) to place a visible marker for each player, whether local or remote, within the 3D map (CoopGpsMarkerController.cs, line 178). In this way, the synchronised GPS status is converted into a spatial position that can be navigated and utilised by the rest of the game's systems.

The same marker controller also links geolocation to the players' visual profiles. To do this, it relies on 'CoopPlayerProfileSync.cs' (line 9), which synchronises each client's 'avatarId' and display name over the network using another 'NetworkList' in 'CoopPlayerProfileSync.cs' (line 15). The local profile is sent to the server using SubmitLocalProfile() and SubmitProfileServerRpc(...) in CoopPlayerProfileSync.cs (line 81). Subsequently, 'CoopGpsMarkerController' queries these profiles to determine which appearance each marker should use via 'ResolveDesiredAvatarId(...)' in 'CoopGpsMarkerController.cs' (line 546). In this way, a player's remote position is shared not only as coordinates, but also as a visual identity within the co-operative map.

The relationship between geolocation and access to missions is reflected in the map's zone system. Each zone is described by 'CoopMinigameZoneDefinition.cs' (line 5), which defines which mission it represents, its display name, the maximum acceptable GPS accuracy threshold, and the spatial tolerance considered valid for being 'within' the zone. CoopMinigameZoneTriggerController.cs (line 11) works with these definitions, periodically checking whether all registered players are simultaneously within the same valid zone (CoopMinigameZoneTriggerController.cs, line 99). To do this, it does not use latitude and longitude directly, but instead queries CoopGpsStateSync for each player's latest status and CoopGpsMarkerController for their position, already projected into the Unity world, using TryGetMarkerWorldPosition(...) in CoopMinigameZoneTriggerController.cs (line 162). If all participants meet the spatial condition and maintain that position during a countdown, the controller requests the CoopSessionCoordinator to load the corresponding minigame using StartMiniGame(...) in CoopMinigameZoneTriggerController.cs (line 158). This is one of the most important relationships in the architecture: the GPS feeds the co-op system and, via the map, acts as a condition for accessing the playable content.

The global session progress layer is located in CoopSessionProgressSync.cs (line 9). This component maintains a NetworkList<CoopMinigameProgressNetworkState> that records which missions have been completed, in what order, and with what average score (CoopSessionProgressSync.cs, line 15). The session coordinator uses this to prevent a minigame that has already been completed from being launched again within the same game, and also to decide whether, after a mission, the group should return to the map or proceed to the final summary scene (CoopSessionCoordinator.cs, line 216). The result of each mission is recorded using 'TryRegisterResultServer(...)', ensuring that the overall progress state remains synchronised across all clients (CoopSessionProgressSync.cs, line 93).

The missions themselves follow a common architecture based on inheritance and reusability. The abstract class CooperativeMinigameBase.cs (line 9) defines the standard lifecycle of all missions: tutorial, waiting for players, active phase and results. To do this, it uses several NetworkVariables, including the phase status, the published result and the list of players who have already completed the tutorial CooperativeMinigameBase.cs (line 14). Each specific mission implements 'InitializeMinigameServer()' to set up its internal state and can call 'PublishResultServer(...)' to communicate the result to both the local minigame flow and the global session progress (CooperativeMinigameBase.cs, line 208). Thanks to this common foundation, all minigames share the same behaviour regarding the tutorial, blocking, progression following results and return to the map, without having to reimplement that flow.

A clear example of this pattern is AudioWordConsensusMinigameSession.cs (line 11). This class inherits from CooperativeMinigameBase and adds its own NetworkVariables to control the active round, sound emitter, disabled words, time remaining, errors made and shared message (AudioWordConsensusMinigameSession.cs, line 17). The server initialises the game, generates the round plan, processes client responses via RPCs and, when the mission ends, publishes a 'MinigameResultData' which is returned to the session coordinator and the global progress system (AudioWordConsensusMinigameSession.cs, line 310; AudioWordConsensusMinigameSession.cs, line 560). The rest of the project's missions follow a similar structure: their own configuration, synchronised session, network models, auxiliary services and specific UI.

In addition to the gameplay logic, there are several supporting components that reinforce the architecture. CoopInformationPresenter.cs (line 5) uses the coordinator's slot mapping to display different variants of information depending on the local player. MapSceneEntryGuard.cs (line 7) prevents invalid direct entries to the map outside the co-op flow. CoopMinigameLauncherUIController.cs (line 11) dynamically builds the panel of available minigames based on settings, the current phase and shared progress. Finally, GpsDebugPanelUIController.cs (line 8) provides a technical debugging view that allows real-time verification of the local reading, the latest synchronised GPS status, the ArcGIS projection and the final position of the marker in the world, which is useful for validating the entire chain from the sensor to the spatial representation.

The project's architecture can be summarised as a layered data flow. First, the device obtains a local GPS reading via DeviceGpsService. Next, CoopGpsMarkerController takes that reading and sends it periodically to CoopGpsStateSync, which replicates it to the other clients via NGO. Next, CoopGpsMarkerController and ArcGISMapCoordinateProjector transform this synchronised state into visible markers on the map. These markers then serve as the basis for the spatial logic of CoopMinigameZoneTriggerController, which detects whether all players are within the same zone and, if so, requests the CoopSessionCoordinator to start the corresponding mission. Once inside the mission, the specific session inherits the common workflow from ‘CooperativeMinigameBase‘, publishes its result and returns it to the coordinator via ‘CoopSessionProgressSync‘, thereby closing the loop between the session, geolocation, map and cooperative progress.